

BMATS101 – MATHEMATICS-I (CSE)

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MODULE 1 : CALCULUS

FOUNDATIONS & POLAR COORDINATES

1. Introduction

Calculus is an important branch of mathematics used to study **rates of change**, **motion**, **curves**, and **geometrical behaviour of functions**.

In **Computer Science and Engineering**, calculus is extensively used in:

- Computer graphics and animation
- Image processing
- Curve fitting and shape analysis
- Optimization and numerical computation

Many curves occurring in engineering applications are difficult to represent in Cartesian form. Hence, **polar coordinates** are introduced to simplify the mathematical representation of such curves.

2. Polar Coordinate System

2.1 Definition

In the polar coordinate system, the position of a point is determined by:

- $r \rightarrow$ length of the radius vector (distance from origin)
- $\theta \rightarrow$ angle made by the radius vector with the positive x-axis

A point is represented as:

(r, θ)

2.2 Important Terms

- **Pole:** Fixed point O (origin)
- **Initial line:** Positive x-axis
- **Radius vector:** Line joining pole to the point
- **Vectorial angle (θ):** Angle measured anticlockwise from initial line

2.3 Range of r and θ

- $r \geq 0$
- θ is measured in **radians**
- Positive direction of θ is **anticlockwise**

3. Conversion Between Cartesian and Polar Coordinates

3.1 Polar to Cartesian

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta\end{aligned}$$

3.2 Cartesian to Polar

$$\begin{aligned}r &= \sqrt{x^2 + y^2} \\ \tan \theta &= \frac{y}{x}\end{aligned}$$

4. Polar Curves

4.1 Definition

A curve whose equation is expressed as:

$$r = f(\theta)$$

is called a **polar curve**.

4.2 Examples of Polar Curves

1. Circle with centre at pole

$$r = a$$

2. Circle passing through pole

$$\begin{aligned}r &= 2a \cos \theta \\ r &= 2a \sin \theta\end{aligned}$$

3. Spiral of Archimedes

$$r = a\theta$$

4. Cardioid

$$r = a(1 + \cos \theta)$$

5. Symmetry of Polar Curves

Symmetry helps in simplifying calculations and tracing curves.

5.1 Symmetry about Initial Line

If replacing θ by $-\theta$ leaves the equation unchanged, the curve is symmetric about the initial line.

5.2 Symmetry about Line $\theta = \frac{\pi}{2}$

If replacing θ by $\pi - \theta$ leaves the equation unchanged, the curve is symmetric about the vertical line.

5.3 Symmetry about the Pole

If replacing θ by $\theta + \pi$ leaves the equation unchanged, the curve is symmetric about the pole.

6. Tangent and Radius Vector

- **Radius vector:** Line joining the pole to any point on the curve
- **Tangent:** Line touching the curve at exactly one point

The angle between these two quantities is important in polar calculus.

7. Angle Between Radius Vector and Tangent

Let the polar curve be:

$$r = f(\theta)$$

Let ψ be the angle between the **radius vector** and the **tangent** at any point.

Correct Formula

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

8. Solved Example

Example 1

Find the angle between the radius vector and the tangent for the curve:

$$r = a \sin \theta$$

Solution

Differentiate with respect to θ :

$$\frac{dr}{d\theta} = a \cos \theta$$

Substitute in formula:

$$\tan \psi = \frac{a \sin \theta}{a \cos \theta}$$

$$\begin{aligned}\tan \psi &= \tan \theta \\ \Rightarrow \psi &= \theta\end{aligned}$$

ANGLE BETWEEN POLAR CURVES & ADVANCED POLAR PROBLEMS

1. Angle Between Two Polar Curves

1.1 Concept

When two curves intersect at a point, the **angle between the curves** is defined as the angle between their **tangents at the point of intersection**.

For polar curves, the tangents are analyzed using the angle made by each tangent with the corresponding **radius vector**.

2. Mathematical Setup

Let the two polar curves be:

$$r_1 = f(\theta), r_2 = g(\theta)$$

Suppose the curves intersect at a point corresponding to the same value of θ .

Let:

- ψ_1 = angle between tangent to curve 1 and its radius vector
- ψ_2 = angle between tangent to curve 2 and its radius vector

Then, the angle between the curves ϕ is:

$$\phi = |\psi_1 - \psi_2|$$

3. Formula for Angle Between Two Polar Curves

From Part 1, we know:

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

Hence,

$$\tan \psi_1 = \frac{r_1}{\frac{dr_1}{d\theta}}, \tan \psi_2 = \frac{r_2}{\frac{dr_2}{d\theta}}$$

Using tangent subtraction formula:

$$\tan (\psi_1 - \psi_2) = \frac{\tan \psi_1 - \tan \psi_2}{1 + \tan \psi_1 \tan \psi_2}$$

Substituting:

Final Formula

$$\tan \phi = \left| \frac{r_1 \frac{dr_2}{d\theta} - r_2 \frac{dr_1}{d\theta}}{r_1 r_2 + \frac{dr_1}{d\theta} \frac{dr_2}{d\theta}} \right|$$

4. Important Observations

1. Curves intersect **orthogonally** if:

$$r_1 r_2 + \frac{dr_1}{d\theta} \frac{dr_2}{d\theta} = 0$$

2. Curves intersect **at right angles** when:

$$\phi = \frac{\pi}{2}$$

5. Solved Problems

Example 1

Find the angle of intersection between the curves:

$$r_1 = a \sin \theta, r_2 = a \cos \theta$$

Solution

Step 1: Differentiate

$$\begin{aligned} \frac{dr_1}{d\theta} &= a \cos \theta \\ \frac{dr_2}{d\theta} &= -a \sin \theta \end{aligned}$$

Step 2: Substitute in formula

$$\tan \phi = \left| \frac{(a \sin \theta)(-a \sin \theta) - (a \cos \theta)(a \cos \theta)}{(a \sin \theta)(a \cos \theta) + (a \cos \theta)(-a \sin \theta)} \right|$$

Simplify numerator:

$$-a^2(\sin^2 \theta + \cos^2 \theta) = -a^2$$

Denominator:

$$a^2(\sin \theta \cos \theta - \sin \theta \cos \theta) = 0$$

Thus,

$$\tan \phi = \infty \Rightarrow \phi = \frac{\pi}{2}$$

Answer

The curves intersect **orthogonally**.

6. Orthogonal Trajectories (Introductory Concept)

A family of curves is said to form **orthogonal trajectories** if each curve of one family cuts every curve of the other family at right angles.

This concept is important in:

- Heat flow
- Electric field lines
- Fluid flow

(Full treatment appears later in ODE module)

7. More Solved Problems

Example 2

Find the angle between the curves:

$$r_1 = a\theta, r_2 = a$$

Solution

Differentiate:

$$\frac{dr_1}{d\theta} = a, \frac{dr_2}{d\theta} = 0$$

Substitute:

$$\begin{aligned}\tan \phi &= \left| \frac{(a\theta)(0) - (a)(a)}{(a\theta)(a) + (a)(0)} \right| \\ \tan \phi &= \left| \frac{-a^2}{a^2\theta} \right| = \frac{1}{\theta}\end{aligned}$$

Answer

$$\phi = \tan^{-1} \left(\frac{1}{\theta} \right)$$

10. Practice Problems (Write Fully in Exam)

1. Find angle of intersection of
 $r = a(1 + \cos \theta)$ and $r = a \sin \theta$
2. Show that curves intersect at right angles:
 $r^2 = a^2 \cos 2\theta$ and $r = a \sin \theta$
3. Find angle between $r = \theta$ and $r = \theta^2$

PEDAL EQUATIONS (THEORY, DERIVATIONS & PROBLEMS)

1. Introduction to Pedal Equations

In many engineering and computer-science applications, it is convenient to describe a curve not by its coordinates directly, but by the **distance of the tangent from a fixed point**. Such equations are called **pedal equations**.

Pedal equations are especially useful in:

- Kinematics
- Mechanical motion
- Curve tracking
- Geometry-based algorithms

2. Pedal of a Curve

2.1 Definition

Given a curve and a fixed point called the **pole**, the **pedal** of the curve is defined as:

The **locus of the foot of the perpendicular** drawn from the pole to the tangent of the curve.

3. Important Quantities in Pedal Form

Consider a curve traced in polar coordinates.

Let:

- **O** be the pole
- **P** be a point on the curve
- **PT** be the tangent at point P

Define:

- r = length of radius vector OP
- ψ = angle between radius vector and tangent
- p = length of perpendicular from the pole to the tangent

4. Fundamental Relation in Pedal Equations

From basic trigonometry:

$$p = r \sin \psi$$

This is the **core formula** of pedal equations.

5. Expression for $\sin \psi$

From earlier results (Module 1 – Part 1):

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

Hence,

$$\sin \psi = \frac{\tan \psi}{\sqrt{1 + \tan^2 \psi}}$$

Substitute:

$$\sin \psi = \frac{\frac{r}{\frac{dr}{d\theta}}}{\sqrt{1 + \left(\frac{r}{\frac{dr}{d\theta}}\right)^2}}$$

Simplifying:

$$\sin \psi = \frac{r}{\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}}$$

6. Pedal Equation in Polar Form

Substitute $\sin \psi$ into $p = r \sin \psi$:

$$p = \frac{r^2}{\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}}$$

7. Pedal Equation of a Given Polar Curve

Given a polar curve:

$$r = f(\theta)$$

Steps to find pedal equation:

1. Differentiate r with respect to θ
2. Substitute r and $\frac{dr}{d\theta}$
3. Use formula for p
4. Eliminate θ if required

8. Special Cases

8.1 When the Tangent is Perpendicular to Radius Vector

$$\psi = \frac{\pi}{2} \Rightarrow p = r$$

8.2 When the Tangent Passes Through the Pole

$$p = 0$$

9. Solved Problems

Example 1

Find the pedal equation of the curve:

$$r = a \cos \theta$$

Solution

Differentiate:

$$\frac{dr}{d\theta} = -a \sin \theta$$

Substitute into pedal formula:

$$p = \frac{a^2 \cos^2 \theta}{\sqrt{a^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

Simplify denominator:

$$\sqrt{a^2 (\cos^2 \theta + \sin^2 \theta)} = a$$

Hence,

$$p = a \cos^2 \theta$$

Final Pedal Equation

$$p = a \cos^2 \theta$$

10. Example 2

Find the pedal equation of the curve:

$$r = a \sin \theta$$

Solution

Differentiate:

$$\frac{dr}{d\theta} = a \cos \theta$$

Substitute:

$$p = \frac{a^2 \sin^2 \theta}{\sqrt{a^2 \sin^2 \theta + a^2 \cos^2 \theta}}$$
$$p = a \sin^2 \theta$$

11. Expressing Pedal Equation in Terms of r Only

From:

$$r = a \sin \theta \Rightarrow \sin \theta = \frac{r}{a}$$

Hence:

$$p = a \left(\frac{r}{a} \right)^2$$

$$p = \frac{r^2}{a}$$

12. Example 3

Find the pedal equation of the spiral:

$$r = a\theta$$

Solution

Differentiate:

$$\frac{dr}{d\theta} = a$$

Substitute:

$$p = \frac{a^2 \theta^2}{\sqrt{a^2 \theta^2 + a^2}}$$
$$p = \frac{a \theta^2}{\sqrt{\theta^2 + 1}}$$

15. Practice Problems

1. Find pedal equation of $r = a(1 + \cos \theta)$
2. Find pedal equation of $r = a\theta^2$
3. Find pedal equation of circle $r = 2a \cos \theta$

CURVATURE AND RADIUS OF CURVATURE

1. Introduction to Curvature

In geometry and engineering applications, it is often required to know **how sharply a curve bends** at a given point.

This property of a curve is measured by **curvature**.

Curvature plays an important role in:

- Computer graphics (smooth curve rendering)
- Image boundary detection
- Robotics and motion control
- Trajectory planning

2. Definition of Curvature

2.1 Curvature

Curvature at a point on a curve is defined as:

The **rate of change of the direction of the tangent** with respect to arc length.

Mathematically,

$$\kappa = \frac{d\psi}{ds}$$

where

ψ = angle made by the tangent with the x-axis

s = arc length

3. Radius of Curvature

3.1 Definition

The **radius of curvature** ρ is defined as the reciprocal of curvature.

$$\rho = \frac{1}{\kappa}$$

Thus,

$$\rho = \frac{ds}{d\psi}$$

4. Circle of Curvature

The **circle of curvature** at a point on a curve is the circle which:

- Touches the curve at that point
- Has the same tangent
- Has the same curvature

The radius of this circle is the **radius of curvature**.

5. Centre of Curvature

The **centre of curvature** is the centre of the circle of curvature.

The locus of centres of curvature of a curve is called the **evolute**.

6. Radius of Curvature in Cartesian Form

6.1 Equation of the Curve

Let the curve be given by:

$$y = f(x)$$

6.2 Derivation

Slope of the curve:

$$\frac{dy}{dx}$$

Angle made by tangent with x-axis:

$$\tan \psi = \frac{dy}{dx}$$

Differentiate w.r.t x :

$$\sec^2 \psi \cdot \frac{d\psi}{dx} = \frac{d^2y}{dx^2}$$

$$\frac{d\psi}{dx} = \frac{\frac{d^2y}{dx^2}}{1 + \left(\frac{dy}{dx}\right)^2}$$

Arc length element:

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Using:

$$\kappa = \frac{d\psi}{ds} = \frac{d\psi/dx}{ds/dx}$$

$$\kappa = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}$$

6.3 Radius of Curvature (Cartesian Form)

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

7. Radius of Curvature in Parametric Form

7.1 Parametric Equations

$$x = f(t), y = g(t)$$

7.2 Required Derivatives

$$\frac{dx}{dt}, \frac{dy}{dt}$$

$$\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}$$

7.3 Radius of Curvature Formula

$$\rho = \frac{\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right]^{3/2}}{\left|\frac{dx}{dt} \frac{d^2y}{dt^2} - \frac{dy}{dt} \frac{d^2x}{dt^2}\right|}$$

8. Radius of Curvature in Polar Form

8.1 Polar Curve

$$r = f(\theta)$$

8.2 Required Quantities

$$\frac{dr}{d\theta}, \frac{d^2r}{d\theta^2}$$

8.3 Radius of Curvature Formula (Polar Form)

$$\rho = \frac{\left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}}{r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2r}{d\theta^2}}$$

9. Radius of Curvature in Pedal Form

Let:

- p = length of perpendicular from pole to tangent
- r = radius vector

Then,

$$\rho = r \frac{dr}{dp}$$

10. Important Remarks

- Curvature depends on **second derivatives**
- Sharp bend $\rightarrow$ small ρ
- Flat curve $\rightarrow$ large ρ

11. Solved Examples

Example 1

Find the radius of curvature of $y = x^2$ at $x = 1$.

Solution

$$\begin{aligned} \frac{dy}{dx} &= 2x \Rightarrow \frac{dy}{dx} = 2 \\ \frac{d^2y}{dx^2} &= 2 \end{aligned}$$

Substitute:

$$\rho = \frac{(1+4)^{3/2}}{2} = \frac{5^{3/2}}{2}$$

Example 2

Find the radius of curvature of the circle:

$$x^2 + y^2 = a^2$$

Solution

A circle has constant radius of curvature.

$$\rho = a$$

SOLVED PROBLEMS, APPLICATIONS & VTU EXAM PREPARATION

1. Large Collection of Solved Problems

Problem 1 (Angle between radius vector and tangent)

Find the angle between the radius vector and the tangent for the curve:

$$r = a(1 + \cos \theta)$$

Solution

Differentiate with respect to θ :

$$\frac{dr}{d\theta} = -a \sin \theta$$

Formula:

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

Substitute:

$$\begin{aligned} \tan \psi &= \frac{a(1 + \cos \theta)}{-a \sin \theta} = -\frac{1 + \cos \theta}{\sin \theta} \\ \Rightarrow \psi &= \tan^{-1} \left(-\frac{1 + \cos \theta}{\sin \theta} \right) \end{aligned}$$

Problem 2 (Angle between two polar curves)

Find the angle of intersection of:

$$r_1 = a \sin \theta, r_2 = a(1 + \cos \theta)$$

Solution

Differentiate:

$$\frac{dr_1}{d\theta} = a \cos \theta, \frac{dr_2}{d\theta} = -a \sin \theta$$

Formula:

$$\tan \phi = \left| \frac{r_1 \frac{dr_2}{d\theta} - r_2 \frac{dr_1}{d\theta}}{r_1 r_2 + \frac{dr_1}{d\theta} \frac{dr_2}{d\theta}} \right|$$

Substitute and simplify:

$$\tan \phi = \left| \frac{a \sin \theta (-a \sin \theta) - a(1 + \cos \theta)(a \cos \theta)}{a \sin \theta \cdot a(1 + \cos \theta) + a \cos \theta (-a \sin \theta)} \right|$$

Final simplification gives the required angle.

Problem 3 (Pedal Equation)

Find the pedal equation of:

$$r = a(1 + \cos \theta)$$

Solution

$$\frac{dr}{d\theta} = -a \sin \theta$$

Pedal formula:

$$p = \frac{r^2}{\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}}$$

Substitute:

$$p = \frac{a^2(1 + \cos \theta)^2}{\sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta}}$$

Simplify denominator using trigonometric identities.

Problem 4 (Radius of Curvature – Cartesian)

Find the radius of curvature of:

$$y = x^3 \text{ at } x = 1$$

Solution

$$\frac{dy}{dx} = 3x^2 \Rightarrow 3$$

$$\frac{d^2y}{dx^2} = 6x \Rightarrow 6$$

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|} = \frac{(1+9)^{3/2}}{6} = \frac{10^{3/2}}{6}$$

Problem 5 (Radius of Curvature – Polar)

Find the radius of curvature of:

$$r = a \sin \theta \text{ at } \theta = \frac{\pi}{2}$$

Solution

$$\frac{dr}{d\theta} = a \cos \theta \Rightarrow 0$$

$$\frac{d^2r}{d\theta^2} = -a \sin \theta \Rightarrow -a$$

Using polar formula:

$$\rho = \frac{(r^2 + (dr/d\theta)^2)^{3/2}}{r^2 + 2(dr/d\theta)^2 - r(d^2r/d\theta^2)}$$

Substitute values to obtain final result.

2. Applications (As Required by Syllabus)

2.1 Applications in Computer Graphics

- Drawing smooth curves using curvature continuity
- Path animation in games
- Bézier and spline curves
- Boundary modeling

Curvature controls **smoothness** of visual objects.

2.2 Applications in Image Processing

- Edge detection
- Shape recognition
- Image segmentation
- Object boundary smoothing

High curvature regions often represent **corners or edges**.

3. Self-Study Topics (Expanded)

3.1 Centre of Curvature

The point at which the centre of the osculating circle lies at a given point of the curve.

3.2 Circle of Curvature

A circle that:

- Touches the curve at one point

- Has same tangent and curvature

3.3 Evolutes

The **locus of centres of curvature** of a given curve.

3.4 Involututes

Curves generated by unwinding a taut string from another curve.

4. VTU Model Questions (Very High Probability)

1. Derive radius of curvature in Cartesian form
2. Find angle between two polar curves
3. Obtain pedal equation of a given polar curve
4. Compute radius of curvature at a given point
5. Applications of curvature in computer graphics

5. One-Day Revision Formula Sheet

Angle between radius vector and tangent

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

Angle between two polar curves

$$\tan \phi = \left| \frac{r_1 \frac{dr_2}{d\theta} - r_2 \frac{dr_1}{d\theta}}{r_1 r_2 + \frac{dr_1}{d\theta} \frac{dr_2}{d\theta}} \right|$$

Pedal length

$$p = r \sin \psi = \frac{r^2}{\sqrt{r^2 + (dr/d\theta)^2}}$$

Radius of curvature (Cartesian)

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$$

Radius of curvature (Polar)

$$\rho = \frac{[r^2 + (dr/d\theta)^2]^{3/2}}{r^2 + 2(dr/d\theta)^2 - r(d^2r/d\theta^2)}$$